

An Uncertainty-aware Resilience Micro-agent for Causal Observability in the Computing Continuum

Suvi De Silva^a, Alfreds Lapkovskis^{a,*}, Sasu Tarkoma^b, Praveen Kumar Donta^a

^aDepartment of Computer and Systems Sciences, Stockholm University, Stockholm 164 25, Sweden.

^bDepartment of Computer Science, University of Helsinki, Helsinki 00000, Finland.

Abstract

The Computing Continuum integrates resource-constrained edge devices, fog nodes, and cloud resources into a unified infrastructure supporting latency-sensitive workloads such as live CCTV monitoring. Grey failures—ambiguous degradations like network congestion—frequently cascade through these heterogeneous environments, simultaneously violating multiple service-level objectives (SLOs) including frames-per-second, end-to-end delay, and throughput. Traditional resilience mechanisms respond by blindly restarting services or adjusting thresholds without structurally identifying the root cause, often executing destructive actions that exacerbate the original fault. This paper presents AURORA (An Uncertainty-awaRe Resilience MicrO-agent for CaUsal Observability in the Computing Continuum Architecture), a framework of lightweight, parallel micro-agents that diagnose and resolve anomalies through causal inference and bounded uncertainty directly on edge hardware. AURORA introduces a dual-gated safety mechanism combining an explicit posterior certainty threshold (≥ 0.85) with a Variational Free Energy (VFE) bound, ensuring healing actions execute only when the inferred root cause is confirmed with high confidence and projected epistemic surprise remains within safe limits. When either gate fails, the system engages in intentional abstention—formally choosing inaction over blind execution. Root-cause identification leverages Judea Pearl’s causal do-calculus over learned Bayesian Network structures, while a functional continuum bridge enables structured edge-to-fog handoff when localized recovery is probabilistically unviable. Experimental evaluations across 1,350 controlled fault-injection trials demonstrate that AURORA achieves 62.7% repair accuracy while maintaining a 0.0% destructive action rate, compared to 33.3% for both rule-based and un-gated Active Inference baselines, establishing the Abstention Paradigm for safe continuum resilience.

Keywords: Active Inference, Computing Continuum, Micro-agent, Resilience, Uncertainty

1. Introduction

The landscape of distributed computing has undergone a fundamental transformation over the past two decades. The traditional centralized cloud computing model, while exceptionally capable for heavy batch processing, large-scale data warehousing, and elastic web services, has proven structurally inadequate for the growing class of latency sensitive, bandwidth-intensive real time applications that modern industry demands [1, 2]. Recognizing this limitation, the research community introduced Edge Computing—a paradigm that physically relocates computational resources closer to the data sources themselves [3]. However, edge nodes alone cannot handle complex global coordination or large-scale model training. This realization gave rise to the modern Computing Continuum: a unified, heterogeneous infrastructure that seamlessly integrates resource constrained edge devices (such as Raspberry Pi cameras and IoT sensor arrays), intermediate fog nodes capable

of cluster-level orchestration, and centralized cloud data centers into a single cohesive abstraction [1]. This paradigm has enabled transformative applications across industrial IoT, autonomous vehicle coordination, smart city infrastructure, and critically for this work real time video surveillance systems where live CCTV streams must be processed with stringent quality guarantees. Nevertheless, the very heterogeneity that gives the Computing Continuum its power also introduces severe operational challenges. The extreme diversity in hardware capabilities, network topologies, and software stacks across tiers means that a single anomalous event—such as a sudden network bandwidth drop, a memory leak in an edge tier Docker container, or an unexpected CPU spike on a constrained Raspberry Pi controller can cascade unpredictably across the continuum. In a live CCTV monitoring scenario, such a grey failure can simultaneously violate multiple service level objectives (SLOs): the frames per second (FPS) drops below 30, the end to end processing delay exceeds 33 milliseconds, and the network throughput falls below the required 1.6 MB/s threshold [4]. These cascading, ambiguous degradations—termed grey failures because they do not produce clean error signals—make system maintenance, fault isolation, and downtime avoidance exceptionally difficult in production continuum deployments.

Several promising approaches have emerged in recent lit-

*Corresponding author

Email addresses: email (Suvi De Silva),
alfreds.lapkovskis@dsv.su.se (Alfreds Lapkovskis),
sasu.tarkoma@helsinki.fi (Sasu Tarkoma), praveen@dsv.su.se
(Praveen Kumar Donta)

erature to address resilience challenges within the Computing Continuum. Active Inference (AIF), grounded in Karl Friston's Free-Energy Principle [5], has demonstrated considerable viability as a cognitive framework for causal diagnosis and adaptive load rebalancing. Sedlak et al. [4] showed that Bayesian Network Structure Learning (BNSL) combined with Markov blankets enables edge devices to develop localized causal understanding of SLO enforcement, requiring only ten training rounds to recover compliance from 22% to 89% in a video-streaming use case. This foundational work was subsequently extended to adaptive stream processing for edge-device bottleneck detection [6]. Moving beyond reactive SLO adjustments, Donta et al. [7] introduced the PAIR-Agent framework, which constructs causal fault graphs from operational logs and manages epistemic uncertainties via Markov blankets, confirming the computational suitability of Active Inference even for severely constrained edge hardware with low memory and CPU footprints. In parallel, decentralized multi-agent architectures such as the Symphony framework [8] have validated the efficacy of distributing lightweight coordination tasks across edge hardware to bypass single node processing limits. However, a critical synthesis of these advances reveals fundamental shortcomings that prevent their safe adoption in production environments. First, existing Active Inference frameworks for the continuum operate primarily as single-agent logical abstractions—they reason causally but do so within a monolithic process, without parallelized deployment across physically distributed edge nodes. Second, while multi-agent architectures like Symphony demonstrate effective parallel coordination, they do not incorporate any formal causal diagnostic model or uncertainty quantification into their decision logic. Third, and most critically, no existing framework in the literature enforces explicit, mathematically bounded safety thresholds such as a posterior certainty gate combined with a Variational Free Energy (VFE) limit—strictly prior to executing physical mitigation actions. This means that current systems, whether rule-based heuristics or probabilistic AIF agents, routinely execute healing actions based on highly uncertain or ambiguous diagnoses. As our experimental results will demonstrate, this unbounded execution directly leads to a 33.3% destructive action rate—interventions that actively exacerbate the original failure rather than resolving it.

Third paragraph start with main challenges addressed in this paper. Introducing Micro-agents, using active inference VFE etc... Then link to the proposed work and refer abbreviation (AURORA) here. By the end of paragraph, list of contributions are as follows:

- Contribution 1
- contribution 2
- Contribution 3

One or two line how does experiments are conducted, which performance metrics are evaluated.

Here, Reamining sections of this paper. .. Section2 ... Section 3

2. Preliminaries

Give brief discussion about active inference, Bayesian network or any other algorithms used in the proposed work from existing.. Broadly.

3. Related Works

[7, 9, 10, 11] Disucss at least 10 works here. If possible, add a table and summarize benefits, methods used, and challenges or limitations..

4. Proposed AURORA

This section might organized as [9, 10]

- Add overview figure,
- split method in to different sections,
- Prefer to add Pseudo code (algorithms) and explain

5. Results and Discussion

5.1. Simulation setup

5.2. Datasets, if any

5.3. Baselines

5.4. Numerical Results

5.4.1. Metric 1

5.4.2. Metric 2

5.4.3. Metric 3

5.4.4. Accuracy

5.4.5. Mean time to recovery

5.5. Discussion and Summary

6. Conclusion and Future Work

Acknowledgment

To be added later

References

- [1] S. Dustdar, V. C. Pujol, P. K. Donta, On distributed computing continuum systems, *IEEE Transactions on Knowledge and Data Engineering* 35 (4) (2022) 4092–4105.
- [2] A. Lapkovskis, B. Sedlak, S. Magnússon, S. Dustdar, P. K. Donta, Benchmarking dynamic slo compliance in distributed computing continuum systems, in: *2025 IEEE International Conference on Edge Computing and Communications (EDGE)*, IEEE, 2025, pp. 93–102.
- [3] S. Dustdar, S. Nastic, O. Scekic, Edge intelligence—a research roadmap, *IEEE Internet of Things Journal* (2020).
- [4] B. Sedlak, P. K. Donta, V. Casamayor Pujol, S. Dustdar, Equilibrium in the computing continuum through active inference, *Future Generation Computer Systems* 160 (2024) 92–108. doi:10.1016/j.future.2024.05.056.
- [5] K. Friston, The free-energy principle: A unified brain theory?, *Nature Reviews Neuroscience* 11 (2) (2010) 127–138.
- [6] B. Sedlak, V. Casamayor Pujol, A. Morichetta, P. K. Donta, S. Dustdar, Adaptive stream processing on edge devices through active inference, *Evolving Systems* 16 (4) (2025) 130. doi:10.1007/s12530-025-09753-2.

- [7] P. K. Donta, A. Lapkovskis, E. Mingozzi, S. Dustdar, Resilient by design—active inference for distributed continuum intelligence, arXiv preprint arXiv:2511.07202 (2025).
- [8] J. Wang, et al., Symphony: A decentralized multi-agent framework for scalable collective intelligence, arXiv preprint arXiv:2508.20019 (2025).
- [9] A. Saleh, P. K. Donta, R. Morabito, S. Tarkoma, A. Lindgren, Q. Zhang, S. Dustdar, S. Pirttikangas, L. Lovén, Bio-inspired agentic self-healing framework for resilient distributed computing continuum systems, arXiv preprint arXiv:2601.00339 (2026).
- [10] P. Ye, A. Lapkovskis, A. Saleh, Q. Zhang, P. K. Donta, Nesy-edge: Neuro-symbolic trustworthy self-healing in the computing continuum, arXiv preprint arXiv:2603.21145 (2026).
- [11] P. K. Donta, B. Sedlak, V. Casamayor Pujol, S. Dustdar, Governance and sustainability of distributed continuum systems: A big data approach, *Journal of Big Data* 10 (1) (2023) 53.